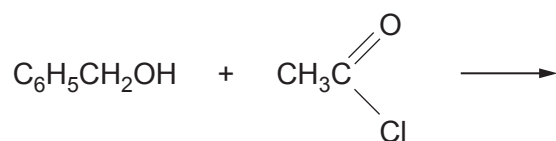


SECTION AAnswer **all** questions.

1. Phenylmethyl ethanoate, used in perfumery, is made by reacting phenylmethanol and ethanoyl chloride.

Complete the equation for this reaction, showing the structure of the organic product. [1]



2. An azo dye has a maximum absorption of visible light at 403 nm.

(a) Calculate the frequency corresponding to this wavelength. [1]

Frequency = Hz

(b) Explain why this dye is yellow in white light. [1]

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